

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE FOR SYNTHESIS, CONTROL, AND PROCESSING

1. ABSTRACT

This report synthesizes a patent set relevant to low-acrylonitrile nitrile butadiene rubber (NBR) and closely related elastomer-processing technologies. The core target is acrylonitrile-butadiene copolymer systems in which acrylonitrile is present at the lower end of the NBR operating window, typically to balance oil resistance, flexibility, processability, and vulcanization behavior. The retained primary patents span classic NBR synthesis control, later controlled-radical emulsion methods, and downstream compounding/process strategies that affect how low-ACN NBR is used in molded or functional rubber articles.

The oldest NBR-specific disclosure in the set, US3862274A, teaches solution, emulsion, or suspension polymerization of nitrile polymers using cycloaliphatic mercaptans as molecular-weight modifiers, with free-radical initiators such as azo, peroxy, or redox systems and polymerization temperatures of about 0 to 100 °C. US5807941A advances NBR process control by copolymerizing acrylonitrile with butadiene in the presence of a free-radical initiator and a long-chain alkylthiol modifier, with the modifier split between the initial monomer charge and a later addition during polymerization when conversion reaches 20 to 70%. This staged chain-transfer strategy is explicitly aimed at improving high-temperature, short-time vulcanizability, processability, and suppression of high-molecular-weight fractions.

The remaining patents are not direct NBR synthesis patents but provide important context for low-ACN NBR applications and process integration. US7592409B2 and EP2345679B1 disclose controlled/living emulsion or aqueous graft polymerization concepts using in-situ emulsifier formation or water-tolerant control agents, illustrating modern approaches to aqueous polymer architecture control. US11130832B2 emphasizes continuous polymerization for narrow molecular-weight distribution and improved processability in conjugated diene polymers. US10731020B2 and US20220372286A1/US10308807B2 focus on compounding and degradable elastomer composites, respectively, where NBR or nitrile rubber can serve as a base elastomer in molded or dissolvable structures. EP3575345B1 and US9228041B2 are outside NBR synthesis but reinforce the broader emulsion-polymerization and latex-processing toolkit, especially coagulation, additive incorporation, and aqueous formulation control.

Overall, the evidence favors low-ACN NBR made by emulsion polymerization with carefully staged monomer and chain-transfer addition, moderate temperatures, and initiator systems that preserve conversion while limiting gel and high-MW tails. Post-polymerization handling and compounding

are equally important for final performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 2

- Patent Number: US5807941A
- Patent Title: Unsaturated nitrile-conjugated diene copolymer process for producing same and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1998
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 3

- Patent Number: US3862274A
- Patent Title: Preparation of nitrile polymers with cycloaliphatic mercaptans as ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1975
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 4

- Patent Number: EP2345679B1
- Patent Title: Method for producing graft copolymer, graft copolymer obtained by the method, ...

- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 5

- Patent Number: EP3575345B1
- Patent Title: Method for producing acrylic rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 6

- Patent Number: US9228041B2
- Patent Title: Compositions with pH responsive copolymer containing MAEP and/or MAHP and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 7

- Patent Number: US20220372286A1
- Patent Title: Dissolvable rubber
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 8

- Patent Number: US10731020B2
- Patent Title: Rubber composition for vulcanization molding, process for manufacturing the ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 9

- Patent Number: US11130832B2
- Patent Title: Modified conjugated diene-based polymer and rubber composition including the ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 10

- Patent Number: US10308807B2
- Patent Title: Dissolvable rubber
- Assignee: Not disclosed
- Jurisdiction: US

- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US3862274A vs US5807941A: both use free-radical polymerization of acrylonitrile/butadiene-type systems, but US3862274A is broader (solution, emulsion, suspension; nitrile polymers with optional comonomers and rubbery graft substrates), while US5807941A is NBR-specific and emphasizes staged addition of an alkylthiol molecular-weight modifier during polymerization.
- US3862274A discloses cycloaliphatic mercaptans such as cyclohexanethiol, cyclododecanethiol, and related dithiols; US5807941A instead requires a long-chain alkylthiol compound with 12 to 16 carbon atoms, at least three tertiary carbon atoms, and sulfur directly bound to at least one tertiary carbon atom.
- US5807941A explicitly states that 10 to 95% of the molecular-weight modifier is added before copolymerization and the remainder is added when conversion reaches 20 to 70%; this staged feed strategy is not specified in US3862274A.
- US3862274A gives a broad polymerization temperature range of about 0 to 100 °C and initiator classes including azo, peroxy, and redox systems; US5807941A's extracted text does not provide a comparable temperature window in the retained passages.
- US7592409B2 and EP2345679B1 both use aqueous/emulsion controlled-polymerization concepts, but they are architecture-control patents rather than NBR synthesis patents; US7592409B2 emphasizes in-situ emulsifier formation with controlled free-radical agents, while

EP2345679B1 uses living radical graft polymerization in water with water-inert control agents such as organotellurium compounds, ATRP agents, RAFT agents, iniferters, or organoiodine compounds.

- US11130832B2 differs from the older NBR patents by focusing on continuous polymerization to obtain a unimodal molecular-weight distribution and PDI below 1.7, highlighting processability and viscoelasticity rather than specific NBR monomer ratios or chain-transfer chemistry.
- US10731020B2, US20220372286A1, and US10308807B2 are downstream elastomer-processing patents rather than NBR synthesis disclosures; they show NBR/nitrile rubber as a base rubber in vulcanization molding or degradable composites, but do not define acrylonitrile content or polymerization recipes in the retained text.
- EP3575345B1 and US9228041B2 are formulation/process patents centered on emulsion polymerization and coagulation/additive handling; they reinforce aqueous processing trends but do not provide NBR-specific monomer ratios or low-ACN synthesis conditions in the retained passages.
- Recurring approach across the NBR-relevant patents: emulsion-based free-radical polymerization with chain-transfer control and post-polymerization processing. Historical patents rely on mercaptan modifiers; newer patents shift toward controlled/living aqueous polymerization, continuous processing, and formulation-level control.
- The main difference between older and newer approaches is the control philosophy: older NBR patents tune molecular weight and vulcanization behavior through thiol feed strategy, while newer aqueous patents emphasize architecture control, narrow dispersity, and process integration in emulsion or continuous systems.

4. CONCLUSION

The patent evidence indicates that low-acrylonitrile NBR is best synthesized by emulsion-based free-radical copolymerization of acrylonitrile and butadiene with deliberate molecular-weight control. The strongest NBR-specific guidance comes from US5807941A, which supports staged addition of a long-chain alkylthiol modifier during polymerization to suppress high-molecular-weight fractions and improve high-temperature, short-time vulcanizability and processability. US3862274A provides broader support for cycloaliphatic mercaptans, free-radical initiators, and polymerization temperatures around 0 to 100 °C across solution, emulsion, or suspension routes.

For low-ACN NBR, the evidence favors maintaining a controlled emulsion process, using a chain-transfer strategy rather than relying only on post-polymerization compounding. The newer patents in the set suggest that continuous or controlled aqueous polymerization can further improve molecular-weight distribution and processability, although they are not NBR-specific in the retained text. Downstream patents reinforce that coagulation, additive timing, and compounding strongly influence final performance.

Recommended synthesis parameters based on the evidence are: emulsion polymerization; free-radical initiation; staged chain-transfer-agent addition; moderate reaction temperatures within the broad 0 to 100 °C range disclosed in the older patent; and process control aimed at limiting high-MW tails while preserving sufficient molecular weight for mechanical strength. For low-ACN embodiments, the most defensible strategy is to keep acrylonitrile at the lower end of the NBR range while using molecular-weight modifiers to preserve processability and vulcanization response.

5. REFERENCES

- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
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- EP2345679B1 | Method for producing graft copolymer, graft copolymer obtained by the method, ... | Not disclosed | EP | 2016 | <https://patents.google.com/patent/EP2345679B1/en>
- EP3575345B1 | Method for producing acrylic rubber | Not disclosed | EP | 2022 | <https://patents.google.com/patent/EP3575345B1/en>
- US9228041B2 | Compositions with pH responsive copolymer containing MAEP and/or MAHP and ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9228041B2/en>
- US20220372286A1 | Dissolvable rubber | Not disclosed | US | 2022 | <https://patents.google.com/patent/US20220372286A1/en>
- US10731020B2 | Rubber composition for vulcanization molding, process for manufacturing the ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10731020B2/en>
- US11130832B2 | Modified conjugated diene-based polymer and rubber composition including the ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11130832B2/en>
- US10308807B2 | Dissolvable rubber | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10308807B2/en>